

PRADNYA SUSHIL SHINDE

Fremont, CA 94536

Phone: (765) 532-9604 | Mail: pshinde1@wpi.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Worcester Polytechnic Institute

Aug 2023 - May 2025

Master of Science in Robotics Engineering (GPA: 4.0/4.0)

Worcester, MA

Relevant Coursework: Computer Vision, Vision-based Robotic Manipulation, Motion Planning, Advanced Robot Navigation

Symbiosis Skills and Professional University

Aug 2019 - May 2023

Bachelor of Technology in Mechatronics Engineering (GPA: 9.128/10.0)

Pune, India

SKILLS

Languages: Python, C++, C

Libraries, Tools, and Software: Linux OS, ROS, ROS (2), Nav2, Gazebo, RViz, Blender, OpenCV, PCL, SLAM Toolbox, NumPy, Matplotlib, Sklearn, PyTorch, Eigen, CUDA, Git, Docker, CMake, Arduino, MATLAB, Visual Basic for Application, Solidworks, Jira, Confluence.

Hardware: Arduino, ESP32, ESP8266, Raspberry Pi, LiDAR

EXPERIENCE

Software Engineer I, Robotics

Jun 2025 - Present

Nightingale Intelligent Systems

Newark, CA

- Contribute to an aerial vehicle software stack, implementing new autonomy features and improving system performance, reliability, and runtime behavior through simulation-driven development.
- Build and maintain a UAV simulation and testing framework from the ground up using ROS and Gazebo, enabling safe validation of autonomy logic before flight testing.

Robotics Software Engineer Intern

Jan 2025 - April 2025

Nightingale Security

Newark, CA

- Optimized UAV path planning by implementing and tuning a **Multi-Heuristic A*** algorithm, improving route efficiency and planner robustness.
- Developed and integrated a simulation **Docker** environment to test UAV functionalities using **ROS** and **Gazebo**.

Graduate Student Researcher

Aug 2024 - May 2025

Worcester Polytechnic Institute

Remote

- Analyzed distributed **SLAM** techniques for agricultural robotics using [Kimera](#), an open-source library for real-time metric-semantic localization and mapping.
- Designed and built a data acquisition pipeline to evaluate the library's applicability and performance in agricultural environments.

Robotics Software Engineer Intern

Aug 2024 - Dec 2024

Celltrio Inc.

Fremont, CA

- Conducted motion analysis of a **7-axis SCARA** robot to accommodate tool transform functionality for precise end-effector manipulation.
- Integrated software fixes to enhance the performance and efficiency of robotic tasks, streamlining overall system functionality.

Robotics Software Engineer Intern

May 2024 - August 2024

National Robotics Engineering Center

Pittsburgh, PA

- Deployed a localization and mapping pipeline for a warehouse mobile manipulator, integrating perception and navigation components.
- Performed **unit testing** and **hardware-in-the-loop** validation to ensure robust performance of the robotic sub-systems.

Robotics Engineer Intern

June 2022 - May 2023

Technodune Pvt. Ltd.

Pune, IN

- Led software testing and embedded systems integration for the "**Wirelessly Controlled Bionic Arm**" and "**Real-Time Robot Localization using Vision**" projects using **Python** and **C/C++**.
- Contributed to developing: A 7DOF robotic manipulator with wireless gesture control and web interface; A real-time robot localization platform with Fiducial markers and Yolo V5, achieving **85%** accuracy in robot pose detection.

PROJECTS

- Einstein Vision: A Perception Pipeline for Autonomous Vehicles** [GitHub](#) | [Python](#) April 2024
- Integrated lane detection, pedestrian pose estimation, road sign recognition, depth estimation, object detection, and scene classification into a unified perception stack, enabling seamless environmental visualization for an autonomous vehicle.
- Motion and Trajectory Tracking of Aerial Vehicle** [GitHub](#) | [Python](#) March 2024
- Implemented state estimation and global positioning algorithms such as **EKF**, **UKF**, **Particle Filter** and **GNSS/INS** to track the motion of an aerial vehicle effectively.
- SfM: Structure from Motion** [GitHub](#) | [Python](#) March 2024
- Engineered a robust a Structure from Motion (SfM) system to reconstruct 3D scenes and estimate camera poses from a monocular camera using images from multiple viewpoints.
- 3D Scene Reconstruction using NeRF** [GitHub](#) | [Python](#) March 2024
- Developed a **NeRF**-based 3D scene reconstruction pipeline using a 5D coordinate system and **MLP** to optimize scene rendering from 2D images, and applied differentiable rendering to minimize residuals between synthesized and ground truth images.
- Zhang's Camera Calibration: Optimizing Distortion Correction** [GitHub](#) | [Python](#) Feb 2024
- Computed camera Intrinsic and Extrinsic Parameters by minimizing image distortion and reprojection error to perform Zhang's **Camera Calibration** method on checkerboard images.
- Homography Estimation: Classical and Deep Learning Methods** [GitHub](#) | [Python](#) Jan 2024
- Designed a panorama stitching algorithm with corner detection, ANMS, feature matching, and **RANSAC**; trained CNN with TensorDLT and STN to predict homography, achieving **3.5 – 5.0%** RMSE.
- Probability-based Edge Detection: A Classical Approach to Boundary Detection** [GitHub](#) | [Python](#) Jan 2024
- Developed a probability-based edge detection algorithm using filter banks, half-disc masks, k-means clustering for texture, brightness, and color features, and integrated with Sobel and Canny baselines for enhanced boundary detection.
- Language Embodied Indoor Navigation** [GitHub](#) | [Python](#) Dec 2023
- Deployed an Object Navigation system enabling robots to interpret spoken language for goal identification and navigate using **RRT*** and **A*** algorithms in the Habitat Simulation Environment, retrieving RGB, depth, and semantic data for sensory feedback.
- Rigid Body Motion Planning** [GitHub](#) | [Python](#) Oct 2023
- Developed and optimized **RRT**, **RRT***, **Informed RRT***, and **Probabilistic Roadmap**(PRM) based path planners tailored to a point robot, a 2D omnidirectional robot, and a 4-link kinematic chain.
- Vision-based Object Grasping for Robotic Manipulation assuming Symmetry** [GitHub](#) | [C++](#) Oct 2023
- Engineered a vision-based grasp pose analysis algorithm for robotic manipulation by processing depth point clouds, estimating symmetry for grasp point computation, using **Gazebo** and **ROS 2** for simulation.

PUBLICATION

Paper Title: “Marathi Sign Language Hand Gesture Recognition Using Accelerometer and 3D Printed Gloves” [Paper](#)
Status: Published in 2022 14th International Conference on Computational Intelligence and Communication Networks (CICN)
Publication Date: January 13, 2023